



Rx Prescription only

Rosuvastatin STELLA
Film-Coated Tablet 10mg

1. **Name of the medicinal product**
Rosuvastatin STELLA Film-Coated Tablet 10mg
2. **Special notice and recommendation**
Keep out of reach of children
Read the package insert carefully before use
3. **Composition**
Each film-coated tablet contains:
Rosuvastatin 10 mg
(as rosuvastatin calcium 10.4 mg)
4. **Pharmaceutical form**
Film-coated tablet.
Light red, round-shaped, film-coated tablet, biconvex, breakable line on one side and plain on the other side.
5. **Indications**
Rosuvastatin is indicated as an adjunct to diet, at least equivalent to the Adult Treatment Panel II (ATP II TLC diet), for the reduction of elevated total cholesterol, LDL-cholesterol, ApoB, the total cholesterol: HDL-cholesterol ratio and triglycerides and for increasing HDL-C, in hyperlipidemic and dyslipidemic conditions, when response to diet and exercise alone has been inadequate including:
Prevention of Cardiovascular Events. In adult patients with an increased risk of atherosclerotic cardiovascular disease based on the presence of cardiovascular disease risk markers such as an elevated hsCRP level, age, hypertension, low HDL-C, smoking or a family history of premature coronary heart disease, Rosuvastatin is indicated to reduce total mortality and the risk of major cardiovascular events (cardiovascular death, stroke, MI, unstable angina, or arterial revascularization).
Rosuvastatin is indicated as an adjunct to diet for the treatment of patients with primary dysbetalipoproteinemia (Type III Hyperlipoproteinemia), Primary hypercholesterolaemia (Type IIa including heterozygous familial hypercholesterolaemia and severe non-familial hypercholesterolaemia), Combined (mixed) dyslipidaemia (Type IIb), Homozygous familial hypercholesterolaemia where Rosuvastatin is used either alone or as an adjunct to diet and other lipid lowering treatment such as apheresis.
Rosuvastatin is indicated as adjunctive therapy to diet to slow the progression of atherosclerosis in adult patients as part of a treatment strategy to lower Total-C and LDL-C to target levels.
Pediatric Patients 10 to 17 years of age with Heterozygous Familial Hypercholesterolemia (HeFH): Adjunct to diet to reduce Total-C, LDL-C and ApoB levels in adolescent boys and girls, who are at least one year post-menarche, 10-17 years of age with heterozygous familial hypercholesterolemia if after an adequate trial of diet therapy the following findings are present: LDL-C > 190 mg/dL or > 160 mg/dL and there is a positive family history of premature cardiovascular disease (CVD) or two or more other CVD risk factors.
6. **Administration and dosage**
Route of administration
Oral
Dosage
Patients should be placed on a standard cholesterol-lowering diet (at least equivalent to the Adult Treatment Panel III (ATP III TLC diet)) before receiving Rosuvastatin, and should continue on this diet during treatment with Rosuvastatin. If appropriate, a program of weight control and physical exercise should be implemented.
Prior to initiating therapy with Rosuvastatin, secondary causes for elevations in plasma lipid levels should be excluded. A lipid profile should also be performed. After initiation or upon titration of Rosuvastatin, lipid levels should be analyzed within 2-4 weeks and the dosage adjusted accordingly.
The usual recommended starting dose of Rosuvastatin is 10 mg once daily. However, initiation of therapy with 5 mg once daily should be considered for special patient populations or patients requiring less aggressive LDL-C reductions. The choice of starting dose should take into account the individual patients' cholesterol level and future cardiovascular risk as well as the potential risk for adverse reactions. Rosuvastatin may be taken in the morning or evening, with or without food. The majority of patients are controlled at the 10 mg dose. However, if necessary, dose adjustments to the next dose level can be made after 4-week intervals. The maximum response is usually achieved within 2-4 weeks and is maintained during chronic therapy. Increasing the dose to 40 mg should be reserved for patients with severe hypercholesterolaemia at high cardiovascular risk (in particular those with familial hypercholesterolaemia), who do not achieve their treatment goal on 20 mg and should only be initiated under specialist supervision (see Special warnings and precautions for use). The physician who elects to use Rosuvastatin at a dose higher than 20 mg should periodically re-evaluate the long term risk/benefit of Rosuvastatin for the individual patient. Rosuvastatin should be prescribed with caution in patients with pre-disposing factors for myopathy/rhabdomyolysis.
The dosage of Rosuvastatin should be individualised according to baseline LDL-C, total-CHDL-C ratio and/or TG levels, the recommended target lipid values and the patient response.
Lipid levels should be monitored periodically and, if necessary, the dose of Rosuvastatin adjusted based on target lipid levels recommended by guidelines.
Dosage in patients with renal insufficiency
The usual dose range applies in patients with mild to moderate renal impairment.
The use of Rosuvastatin Calcium in patients with severe renal impairment is contraindicated.
Dosage in patients with hepatic insufficiency
There was no increase in systemic exposure to Rosuvastatin in subjects with Child-Pugh scores of 7 or below. However, increased systemic exposure has been observed in subjects with Child-Pugh scores of 8 and 9. In these patients an assessment of renal function should be considered. There is no experience in subjects with Child-Pugh scores above 9. Rosuvastatin calcium is contraindicated in patients with active liver disease.
Use in the elderly
The overall frequency of adverse events and types of adverse events were similar in patients above and below 65 years of age. The efficacy of Rosuvastatin in the geriatric population (≥ 65 years of age) was comparable to the efficacy observed in the non-elderly.
Pediatric patients (10 to 17 years of age)
In pediatric patients (10 to 17 years of age) with heterozygous familial hypercholesterolemia the usual dose range of Rosuvastatin is 5-20 mg/day; the maximum recommended dose is 20 mg/day (doses greater than 20 mg have not been studied in this patient population). Doses should be individualized according to the recommended goal of therapy. Adjustments should be made at intervals of 4 weeks or more.
Use in children below 10 years
The safety and effectiveness in children have not been established. In children and adolescents with homozygous familial hypercholesterolemia experience is limited to eight patients (aged 8 years and above).
Dosage on Asian patients
Initiation of Rosuvastatin therapy with 5 mg once daily should be considered for Asian patients. The potential for increased systemic exposures relative to Caucasians is relevant when considering escalation of dose in cases where hypercholesterolaemia is not adequately controlled at doses of 5, 10 or 20 mg once daily.
Genetic polymorphisms
Specific types of genetic polymorphisms are known that can lead to increased Rosuvastatin exposure (see Pharmacokinetic Properties). For patients who are known to have such specific types of polymorphisms, a lower daily dose of Rosuvastatin is recommended.
Dosage in patients with pre-disposing factors to myopathy
The recommended start dose is 5 mg in patients with pre-disposing factors to myopathy.
Concomitant therapy
Rosuvastatin is a substrate of various transporter proteins (e.g. OATP1B1 and BCRP). The risk of myopathy (including rhabdomyolysis) is increased when Rosuvastatin is administered concomitantly with certain medicinal products that may increase the plasma concentration of Rosuvastatin due to interactions with these transporter proteins (e.g. ciclosporin and certain protease inhibitors including combinations of ritonavir with atazanavir, lopinavir, and/or tipranavir).
Whenever possible, alternative medications should be considered, and, if necessary, consider temporarily discontinuing Rosuvastatin therapy. In situations where co-administration of these medicinal
- products with Rosuvastatin is unavoidable, the benefit and the risk of concurrent treatment and Rosuvastatin dosing adjustments should be carefully considered.
Rosuvastatin STELLA Film-Coated Tablet 10mg is not able to deliver all approved dose regimens of Rosuvastatin. Therefore, other approved dosage forms and strengths of Rosuvastatin should be used in such cases.

7. **Contraindications**
Rosuvastatin is contraindicated:
– in patients with hypersensitivity to any component of this product.
– in patients with active liver disease including unexplained, persistent elevations of serum transaminases and any serum transaminase elevation exceeding 3x the upper limit of normal (ULN).
– during pregnancy, while breast-feeding and in women of child-bearing potential not using appropriate contraceptive measures.
– in patients with myopathy.
– in patients receiving concomitant cyclosporine.

8. **Special warnings and precautions for use**
Renal effects
Proteinuria, detected by dipstick testing and mostly tubular in origin, has been observed in patients treated with higher doses of Rosuvastatin, in particular 40 mg, where it was transient or intermittent in most cases. Proteinuria has not been shown to be predictive of acute or progressive renal disease. An assessment of renal function should be considered during routine follow-up of patients treated with a dose of 40 mg.
Renal impairment
Rosuvastatin exposure is not influenced by mild to moderate renal impairment (CLcr ≥30 mL/min/1.73 m²). Exposure to Rosuvastatin is increased to a clinically significant extent in patients with severe renal impairment (CLcr <30 mL/min/1.73 m²) who are not receiving hemodialysis and dose adjustment is required.
Skeletal muscle effects
Gemfibrozil increases the risk of myopathy when given concomitantly with some HMG-CoA reductase inhibitors. Therefore, the combination of Rosuvastatin and gemfibrozil is not recommended. The benefit of further alterations in lipid levels by the combined use of Rosuvastatin with fibrates or niacin should be carefully weighed against the potential risks of such combinations.
Effects on skeletal muscle e.g. myalgia, myopathy and, rarely, rhabdomyolysis have been reported in Rosuvastatin-treated patients with all doses and in particular with doses >20 mg. Very rare cases of rhabdomyolysis have been reported with the use of ezetimibe in combination with HMG-CoA reductase inhibitors. A pharmacodynamic interaction cannot be excluded and caution should be exercised with their combined use. As with other HMG-CoA reductase inhibitors, the reporting rate for rhabdomyolysis associated with Rosuvastatin in post-marketing use is higher at the 40 mg dose.
Creatine Kinase Measurement
Creatine Kinase (CK) should not be measured following strenuous exercise or in the presence of a plausible alternative cause of CK increase which may confound interpretation of the result. If CK levels are significantly elevated at baseline (>5 x ULN) a confirmatory test should be carried out within 5 – 7 days. If the repeat test confirms a baseline CK >5 x ULN, treatment should not be started.
Before Treatment
Rosuvastatin, as with other HMG-CoA reductase inhibitors, should be prescribed with caution in patients with pre-disposing factors for myopathy/rhabdomyolysis. Such factors include:
+ renal impairment
+ hypothyroidism
+ personal or family history of hereditary muscular disorders
+ previous history of muscular toxicity with another HMG-CoA reductase inhibitor or fibrate
+ alcohol abuse
+ age >70 years
+ situations where an increase in plasma levels may occur
+ concomitant use of fibrates.
In such patients the risk of treatment should be considered in relation to possible benefit and clinical monitoring is recommended. If CK levels are significantly elevated at baseline (>5 x ULN) treatment should not be started.
Whilst on Treatment
Patients should be asked to report inoperable muscle pain, weakness or cramps immediately, particularly if associated with malaise or fever. CK levels should be measured in these patients. Therapy should be discontinued if CK levels are markedly elevated (>5 x ULN) or if muscular symptoms are severe and cause daily discomfort (even if CK levels are ≤5 x ULN). If symptoms resolve and CK levels return to normal, then consideration should be given to re-introducing Rosuvastatin or an alternative HMG-CoA reductase inhibitor at the lowest dose with close monitoring. Routine monitoring of CK levels in asymptomatic patients is not warranted. There have been very rare reports of an immune-mediated antineutrophilic myopathy (NMN) during or after treatment with some statins, including Rosuvastatin. NMN is clinically characterised by:
+ persistent proximal muscle weakness and elevated serum creatine kinase, which persist despite discontinuation of statin treatment.
+ muscle biopsy showing necrotizing myopathy without significant inflammation;
+ improvement with immunosuppressive agents.
There was no evidence of increased skeletal muscle effects in the small number of patients dosed with Rosuvastatin and concomitant therapy. However, an increase in the incidence of myositis and myopathy has been seen in patients receiving other HMG-CoA reductase inhibitors together with fibric acid derivatives including gemfibrozil, ciclosporin, nicotinic acid, azole antifungals, protease inhibitors and macrolide antibiotics. Gemfibrozil increases the risk of myopathy when given concomitantly with some HMG-CoA reductase inhibitors. Therefore, the combination of Rosuvastatin and gemfibrozil is not recommended. The benefit of further alterations in lipid levels by the combined use of Rosuvastatin with fibrates or niacin should be carefully weighed against the potential risks of such combinations.
Rosuvastatin should not be used in any patient with an acute, serious condition suggestive of myopathy or predisposing to the development of renal failure secondary to rhabdomyolysis (e.g. sepsis, hypotension, major surgery, trauma, severe metabolic, endocrine and electrolyte disorders; or uncontrolled seizures).

Myasthenia Gravis/Ocular Myasthenia
In few cases, statins have been reported to induce de novo or aggravate pre-existing myasthenia gravis or ocular myasthenia. Rosuvastatin should be discontinued in case of aggravation of symptoms. Recurrences when the same or a different statin was (-re) administered have been reported.

Liver effects
As with other HMG-CoA reductase inhibitors, Rosuvastatin should be used with caution in patients who consume excessive quantities of alcohol and/or have a history of liver disease.
It is recommended that liver function tests be carried out prior to, and 3 months following, the initiation of treatment. Rosuvastatin should be discontinued or the dose reduced if the level of serum transaminases is greater than 3 times the upper limit of normal. The reporting rate for serious hepatic events (consisting mainly of increased hepatic transaminases) in post-marketing use is higher at the 40 mg dose.
In patients with secondary hypercholesterolaemia caused by hypothyroidism or nephrotic syndrome, the underlying disease should be treated prior to initiating therapy with Rosuvastatin.

Race
Pharmacokinetic studies show an increase in exposure in Asian patients compared with Caucasians.

Protease inhibitors
Increased systemic exposure to Rosuvastatin has been observed in patients receiving Rosuvastatin concomitantly with various protease inhibitors in combination with ritonavir. Consideration should be given both to the benefit of lipid lowering by use of Rosuvastatin in HIV patients receiving protease inhibitors and the potential for increased Rosuvastatin plasma concentrations when initiating and up titrating Rosuvastatin doses in patients treated with protease inhibitors. The concomitant use with certain protease inhibitors is not recommended unless the dose of Rosuvastatin is adjusted.

Lactose intolerance
Patients with rare hereditary problems of galactose intolerance, the Lapp lactase deficiency or glucose-galactose maldigestion should not take this medicine.

Interstitial lung disease
Exceptional cases of interstitial lung disease have been reported with some statins, especially with long term therapy. Presenting features can include dyspnoea, non-productive cough and deterioration in general health (fatigue, weight loss and fever). If it is suspected a patient has developed interstitial lung disease, statin therapy should be discontinued.

Diabetes mellitus
Some evidence suggests that statins as a class raise blood glucose and in some patients, at high risk of future diabetes, may produce a level of hyperglycaemia where formal diabetes care is appropriate.

32.5 mm

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This risk, however, is outweighed by the reduction in vascular risk with statins and therefore should not be a reason for stopping statin treatment. Patients at risk (fasting glucose ≥ 5.6 to 6.9 mmol/L, BMI >30 kg/m², raised triglycerides, hypertension) should be monitored both clinically and biochemically according to national guidelines.

Pediatric patients (10 to 17 years of age)

The evaluation of linear growth (height), weight, BMI (body mass index), and secondary characteristics of sexual maturation by Tanner staging in pediatric patients taking Rosuvastatin is limited to a one year period.

9. Pregnancy and lactation

The safety of Rosuvastatin during pregnancy and whilst breast-feeding has not been established. Women of child-bearing potential should use appropriate contraceptive measures.

10. Effects on ability to drive and use machines

Pharmacology testing revealed no evidence of a sedative effect of Rosuvastatin. From the safety profile, Rosuvastatin is not expected to adversely affect the ability to drive or use machines.

11. Drug interactions

Effect of co-administered medicinal products on Rosuvastatin

- Transporter protein inhibitors: Rosuvastatin is a substrate for certain transporter proteins including the hepatic uptake transporter OATP1B1 and efflux transporter BCRP. Concomitant administration of Rosuvastatin with medicinal products that are inhibitors of these transporter proteins may result in increased Rosuvastatin plasma concentrations and an increased risk of myopathy.
 - Ciclosporin: Ciclosporin increased Rosuvastatin exposure and may result in increased risk of myopathy. Therefore, in patients taking ciclosporin, the dose of Rosuvastatin should not exceed 5 mg once daily.
 - Protease inhibitors: Although the exact mechanism of interaction is unknown, concomitant protease inhibitor use may strongly increase Rosuvastatin exposure. Co-administration of 10 mg Rosuvastatin and a combination product of two protease inhibitors (300 mg atazanavir / 100 mg ritonavir) in healthy people was associated with an approximately three-fold and seven-fold increase in Rosuvastatin AUC and C_{max}, respectively. The concomitant use of Rosuvastatin and some protease inhibitor combinations may be considered after careful consideration of Rosuvastatin dose adjustments based on the expected increase in Rosuvastatin exposure.
 - Gemfibrozil and other lipid-lowering products: Concomitant use of Rosuvastatin and gemfibrozil resulted in a 2-fold increase in Rosuvastatin C_{max} and AUC. No pharmacokinetic relevant interaction with fenofibrate is expected, however a pharmacodynamic interaction may occur. Gemfibrozil, fenofibrate, other fibrates and lipid lowering doses (0 or equal to 1g/day) of niacin (nicotinic acid) increase the risk of myopathy when given concomitantly with HMG-CoA reductase inhibitors, probably because they can produce myopathy when given alone. These patients should also start with the 5 mg dose.
 - Ezetimibe: Concomitant use of 10 mg Rosuvastatin and 10 mg ezetimibe resulted in a 1.2- fold increase in AUC of Rosuvastatin in hypercholesterolaemic patients. A pharmacodynamic interaction, in terms of adverse effects, between Rosuvastatin and ezetimibe cannot be ruled out.
 - Antacid: The simultaneous dosing of Rosuvastatin with an antacid suspension containing aluminium and magnesium hydroxide resulted in a decrease in Rosuvastatin plasma concentration of approximately 50%. This effect was mitigated when the antacid was dosed 2 hours after Rosuvastatin. The clinical relevance of this interaction has not been studied.
 - Fusidic Acid: Interaction studies with Rosuvastatin and fusidic acid have not been conducted. As with other statins, muscle related events, including rhabdomyolysis, have been reported with Rosuvastatin and fusidic acid given concurrently. Patients should be closely monitored and temporary suspension of Rosuvastatin treatment may be appropriate.
 - Erythromycin: Concomitant use of Rosuvastatin and erythromycin resulted in a 20% decrease in AUC (0-4) and a 30% decrease in C_{max} of Rosuvastatin. This interaction may be caused by the increase in gut motility caused by erythromycin.
 - Cytochrome P450 enzymes: Rosuvastatin is neither an inhibitor nor an inducer of cytochrome P450 isoenzymes. In addition, Rosuvastatin is a poor substrate for these isoenzymes. Therefore, drug interactions resulting from cytochrome P450-mediated metabolism are not expected. No clinically relevant interactions have been observed between Rosuvastatin and either fluconazole (an inhibitor of CYP2C9 and CYP3A4) or ketonazole (an inhibitor of CYP2A6 and CYP3A4).
 - Interactions requiring Rosuvastatin dose adjustments
- When it is necessary to co-administer Rosuvastatin with other medicinal products known to increase exposure to Rosuvastatin, doses of Rosuvastatin should be adjusted. Start with a 5 mg once daily dose of Rosuvastatin if the expected in exposure (AUC) is approximately 2-fold or higher.
- The maximum daily dose of Rosuvastatin should be adjusted so that the expected Rosuvastatin exposure would not likely exceed that of the recommended maximum daily dose of Rosuvastatin taken without interacting medicinal products. For example, where the recommended dose of Rosuvastatin is 20 mg, the dose of Rosuvastatin taken with a ritonavir/atazanavir combination (3.1-fold increase) should not exceed 5 mg, and the dose of Rosuvastatin taken with gemfibrozil (1.9-fold increase) should not exceed to 10 mg.

Table 3. Effect of co-administered medicinal products on Rosuvastatin exposure (AUC, in order of decreasing magnitude)

Interacting drug dose regimen	Rosuvastatin dose regimen	Change in rosuvastatin AUC*
Cholevidon 75 mg BD to 250 mg BD, 6 months	10 mg QD, 10 days	7.1-fold ↑
Alosetron 300 mg/ritonavir 100 mg QD, 8 days	10 mg, single dose	3.1-fold ↑
Simeprevir 150 mg QD, 7 days	10 mg, single dose	2.8-fold ↑
Lopinavir 400 mg/ritonavir 100 mg BD, 12 days	20 mg QD, 7 days	2.1-fold ↑
Cholevidon 300 mg loading, followed by 75 mg at 24 hours*	20 mg, single dose†	2-fold ↑
Gemfibrozil 600 mg BD, 7 days	80 mg, single dose	1.9-fold ↑
Ethionitrop 75 mg QD, 5 days	10 mg, single dose	1.6-fold ↑
Danavone 500 mg/ritonavir 100 mg BD, 7 days	10 mg QD, 7 days	1.5-fold ↑
Tenavir 500 mg/ritonavir 200 mg BD, 11 days	10 mg, single dose	1.4-fold ↑
Drosedrone 400 mg BD	Not available	1.4-fold ↑
Inositol 200 mg QD, 5 days	10 mg, single dose	1.4-fold ↑*
Ezetimibe 10 mg QD, 14 days	10 mg QD, 14 days	1.2-fold ↑*
Rosuvastatin 700 mg/ritonavir 100 mg BD, 8 days	10 mg, single dose	++
Alosetron 0.3 mg, 7 days	40 mg, 7 days	++
Sildenafil 140 mg BD, 5 days	10 mg, single dose	++
Fenofibrate 150 mg TID, 7 days	20 mg, 7 days	++
Rifampin 400 mg QD, 7 days	20 mg, single dose	++
Ketconazole 200 mg BD, 7 days	80 mg, single dose	++
Fluconazole 200 mg QD, 11 days	80 mg, single dose	++
Erythromycin 500 mg QD, 7 days	80 mg, single dose	20% ↓
Baclofen 30 mg TID, 14 days	20 mg, single dose	47% ↓

*Data given as x-fold change represent a single ratio between co-administration and rosuvastatin alone. Data given as % change represent % difference relative to rosuvastatin alone.

↑ Increase is indicated as "+", decrease as "-".

** The table shows the most significant ratio

QD = once daily; BD = twice daily; TID = three times daily; QID = four times daily

- Other medications: Concurrent use of fibrates may cause severe myositis and myoglobinuria.

Effect of Rosuvastatin on co-administered medicinal products

- Vitamin K antagonists: As with other HMG-CoA reductase inhibitors, the initiation of treatment or dosage up-titration of Rosuvastatin in patients treated concomitantly with vitamin K antagonists (e.g. warfarin or another coumarin anticoagulant) may result in an increase in International Normalised Ratio (INR). Discontinuation or down-titration of Rosuvastatin may result in a decrease in INR. In such situations, appropriate monitoring of INR is desirable.
- Oral contraceptive/hormone replacement therapy (HRT): Concomitant use of Rosuvastatin and an oral contraceptive resulted in an increase in ethinyl estradiol and norgestrel AUC of 26% and 34%, respectively. These increased plasma levels should be considered when selecting oral contraceptive doses. There are no pharmacokinetic data available in patients taking concomitant Rosuvastatin and HRT and therefore a similar effect cannot be excluded. However, the combination has been extensively used in women and was well tolerated.
- Other medicinal products: No clinically relevant interaction with digoxin is expected.
- Paediatric population: The extent of interactions in the paediatric population is not known.

12. Adverse reactions

There have been rare reports of cognitive impairment (e.g. memory loss, forgetfulness, amnesia, memory impairment, confusion) associated with statin use. These issues have been reported for all statins. The reports are generally non-serious and reversible upon statin discontinuation, with variable times to symptom onset (1 day to years) and symptom resolution (median 3 weeks). Increases with HbA1c and fasting blood glucose have been reported with statins. The risk of hyperglycemia, however, is outweighed by the reduction in vascular risk with statins. The adverse reactions seen with Rosuvastatin are generally mild and transient. Adverse reactions listed below are classified according to frequency and system organ class (SOC).

- Blood and lymphatic system disorders: **Rare:** thrombocytopenia.
- Immune system disorders: **Rare:** hypersensitivity reactions including angioedema.
- Endocrine disorders: **Common:** diabetes mellitus [Frequency will depend on the presence or absence of risk factors (fasting blood glucose ≥ 5.6 mmol/L, BMI >30 kg/m², raised triglycerides, history of hypertension)].
- Psychiatric disorders: **Not known:** depression.
- Nervous system disorders: **Common:** headache, dizziness; **Very rare:** polyneuropathy, memory loss; **Not known:** peripheral neuropathy, sleep disturbances (including insomnia and nightmares), myasthenia gravis.
- Eye disorders: **Not known:** ocular myasthenia.
- Respiratory, thoracic and mediastinal disorders: **Not known:** cough, dyspnea.
- Gastrointestinal disorders: **Common:** constipation, nausea, abdominal pain; **Rare:** pancreatitis; **Not known:** diarrhea.
- Hepatobiliary disorders: **Rare:** increased hepatic transaminases; **Very rare:** jaundice, hepatitis.
- Skin and subcutaneous tissue disorders: **Uncommon:** pruritis, rash, urticaria; **Not known:** Stevens-Johnson syndrome.
- Musculoskeletal and connective tissue disorders: **Common:** myalgia; **Rare:** myopathy (including myositis), rhabdomyolysis; **Very rare:** arthralgia; **Not known:** immune-mediated necrotizing myopathy.
- Renal and urinary disorders: **Very rare:** haematuria.
- Reproductive system and breast disorders: **Very rare:** gynaecomastia.
- General disorders and administration site conditions: **Common:** asthenia; **Not known:** oedema.

As with other HMG-CoA reductase inhibitors, the incidence of adverse drug reactions tends to be dose dependent.

Renal effects: Proteinuria, detected by dipstick testing and mostly tubular in origin, has been observed in patients treated with Rosuvastatin. Shifts in urine protein from none or trace to ++ or more were seen in <1% of patients at some time during treatment with 10 and 20 mg, a minor increase in shift from none or trace to + was observed with the 20 mg dose. In most cases, proteinuria decreases or disappears spontaneously on continued therapy. Available data has not identified a causal association between proteinuria and acute or progressive renal disease. Haematuria has been observed in patients treated with Rosuvastatin but clinical data show that the occurrence is low.

Skeletal muscle effects: Effects on skeletal muscle e.g. myalgia, myopathy (including myositis) and, rarely, rhabdomyolysis with and without acute renal failure have been reported in Rosuvastatin-treated patients with all doses and in particular with doses >20 mg. A dose-related increase in CK levels has been observed in patients taking Rosuvastatin; the majority of cases were mild, asymptomatic and transient. If CK levels are elevated ($>5 \times$ ULN), treatment should be discontinued.

Liver effects: As with other HMG-CoA reductase inhibitors, a dose-related increase in transaminases has been observed in a small number of patients taking Rosuvastatin; the majority of cases were mild, asymptomatic and transient.

The following adverse events have been reported with some statins:

+ Sexual dysfunction.

+ Exceptional cases of interstitial lung disease, especially with long term therapy.

+ Tendon disorders, sometimes complicated by rupture.

The reporting rates for rhabdomyolysis, serious renal events and serious hepatic events (consisting mainly of increased hepatic transaminases) is higher at the 40 mg dose.

13. Overdose and management

There is no specific treatment in the event of overdose. In the event of overdose, the patient should be treated symptomatically and supportive measures instituted as required.

Haemodialysis is unlikely to be of benefit.

14. Pharmacodynamic properties

Pharmaco-therapeutic group: HMG-CoA reductase inhibitors

ATC code: C10A A07

Mechanism of action

Rosuvastatin is a selective, potent and competitive inhibitor of HMG-CoA reductase, the rate-limiting enzyme that converts 3-hydroxy-3-methylglutaryl coenzyme A to mevalonate, a precursor of cholesterol. Triglycerides (TG) and cholesterol in the liver are incorporated, with apolipoprotein B (ApoB), into very low density lipoprotein (VLDL) and released into the plasma for delivery to peripheral tissues. VLDL particles are TG-rich. Cholesterol-rich low density lipoprotein (LDL) is formed from VLDL and is cleared primarily through the high affinity LDL receptor in the liver. Rosuvastatin produces its lipid-modifying effects in two ways; it increases the number of hepatic LDL receptors on the cell-surface, enhancing uptake and catabolism of LDL and it inhibits the hepatic synthesis of VLDL, thereby reducing the total number of VLDL and LDL particles.

High density lipoprotein (HDL), which contains ApoA-I is involved, amongst other things, in transport of cholesterol from tissues back to the liver (reverse cholesterol transport).

The involvement of LDL-C in atherosclerosis has been well documented. Epidemiological studies have established that high LDL-C, TG, low HDL-C and ApoB-I have been linked to a higher risk of cardiovascular disease. Intervention studies have shown the benefits on mortality and CV event rates of lowering LDL-C and TG or raising HDL-C. More recent data has linked the beneficial effects of HMG-CoA reductase inhibitors to lowering of non-HDL (i.e. all circulating cholesterol not in HDL) and ApoB or reducing the ApoB/ApoA-I ratio.

15. Pharmacokinetic properties

Absorption: Peak plasma concentrations of Rosuvastatin were reached 3 to 5 hours following oral dosing. Both peak concentration (C_{max}) and area under the plasma concentration-time curve (AUC) increased in approximate proportion to Rosuvastatin dose. The absolute bioavailability of Rosuvastatin is approximately 20%.

Administration of Rosuvastatin with food decreased the rate of drug absorption by 20% as assessed by C_{max}, but there was no effect on the extent of absorption as assessed by AUC.

Plasma concentrations of Rosuvastatin do not differ following evening or morning drug administration.

Significant LDL-C reductions are seen when Rosuvastatin is given with or without food, and regardless of the time of day of drug administration.

Distribution: Mean volume of distribution at steady-state of Rosuvastatin is approximately 124 litres. Rosuvastatin is 98% bound to plasma proteins, mostly albumin. This binding is reversible and independent of plasma concentrations.

Metabolism: Rosuvastatin is not extensively metabolised; approximately 10% of a radiolabelled dose is recovered as metabolite.

The major metabolite is N-desmethyl rosuvastatin, which is formed principally by cytochrome P450 2C9, and in vitro studies have demonstrated that N-desmethyl rosuvastatin has approximately one-sixth to one-half the HMG-CoA reductase inhibitory activity of Rosuvastatin. Overall, greater than 90% of active plasma HMG-CoA reductase inhibitory activity is accounted for by Rosuvastatin.

Excretion: Following oral administration, Rosuvastatin and its metabolite are primarily excreted in the faeces (90%). The elimination half-life (t_{1/2}) of Rosuvastatin is approximately 19 hours.

After an intravenous dose, approximately 28% of total body clearance was via the renal route and 72% by the hepatic route.

16. Packaging

B blister of 10 tablets. Box of 3 blisters.

17. Storage condition, shelf-life

Do not store above 30°C. Store in the original package.

The expiry date of this pack is printed on the box. Do not use this pack after this date.

18. Name, address of manufacturer

STELLAPHARM J.V. CO. LTD. - BRANCH 1
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STELLA

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